HYUNDAI-ROTEM (H-R) SUBMITTAL REVIEW FORM

H-R Submittal Reference:	HRU-TRN-000170 Design Data Package: #10 (PDR)	**Date Submitted:** 13th May 2011	**DTS Review #** DTS-HRU-R0113

Doc No.	Title	Rev	CDRL Ref.	Spec Ref
See Below	Communications & CCTV PDR Design Package (Detailed Below)	1	9-16.R15	No Specific Reference

DTS/DTP Submittal Review Comments :

The following revised and/or additional documents were submitted for review as part of the Communications & CCTV PDR Design Package in response to DTS initial review.

- HR Document REDU107542, Rev. 1 – Video Surveillance Camera Placement & Coverage Analysis
- HR Document REDE113108, Rev. 1 – Technical Description for CCTV System
- HR Document REDU113112, Rev. 1 – Technical Description for Communications System

All DTS comments submitted with DTS-HRU-R0080 have been addressed by HR and can be considered closed except for the following:

Comment 2.2 – As emailed to Mr. Leo Jeon on June 23, DTS & RTD still have concerns about CCTV camera coverage in the following locations: F-end right side seat adjacent to the electrical locker and at the side door entry areas. An alternative CCTV camera arrangement was sent in the same email for HR and Inter-M's consideration. Please evaluate and determine the optimal camera layout to ensure "complete (100%) coverage of the passenger area including boarding/door areas".

☑ *See continuation sheet*

Approval Status *	A	B	C	IO
PDR		X		
FDR				

Reviewer's Signature:		**Review Date:**	27th June 2011

**** Key to Approval Status Codes:***

A = Proceed
B = Revise and Resubmit. Work may proceed subject to incorporation of changes indicated.
C = Do not proceed. Change as noted and resubmit.
IO = Data accepted for information only.

DTS/DTP Submittal Review Comments Continued:

Comment 2.7 – Please proceed with the CCTV monitor darkening while at speed concept as discussed at the PDR. Please provide an override function to allow the operator to monitor the CCTV cameras while at speed.

The following action item remains open from the PDR meeting:

- RS-MM-036-25 (DTS action)

The following are new DTS comments:

2.10 General – Although it was not labeled as an action item from the PDR meeting, please provide the user configurable decisions matrix for DTS/DTO evaluation.

2.11 HR Document REDE113112, Page 11 – As discussed at the PDR, the intent train crew to emergency responder communications requirement in APTA SS-PS-001-98 is to allow emergency responders to communicate to the crew and passengers from outside the vehicle. It was suggested at the PDR to add an external jack that the emergency personnel could plug into that would immediately connect them to the vehicle communications system. Please investigate whether this function can be added to the vehicle.

2.12 HR Document REDE113112, Section 3.6 – As discussed at the PDR, please ensure that it is not possible to completely mute the cab communications speaker.

2.13 HR Document REDE113112, Section 3.8 – As discussed at the PDR, please do not broadcast Passenger Emergency Intercom activations throughout the passenger speakers.

2.14 HR Document REDE113112, Section 3.8 – As discussed at the PDR, please implement the functionality to allow the OCC to monitor Passenger Emergency Intercom communications with the operator. Please also provide a means for the operator or the OCC to be notified when the other party has taken over the emergency communications.

2.15 HR Document REDE113108 – As discussed at the PDR meeting, please ensure the automatic viewing of the exterior cameras when the vehicle is stopped can be overridden.

2.16 HR Document REDU107542, Attachment – Please ensure that any windshield visors are taken into consideration when locating the cab forward facing camera.

DTS/DTP Submittal Review Comments Continued:

2.17 Please confirm that the Ethernet based communications backbone should run the full length of the train, not just the communications cabinet suggested in the PDR package. HR should integrate the onboard communications system to utilize the trainline fully and take advantage of the flexibility offered by IP based communications.

2.18 Can PEI calls be routed to the operator's hand portable radio in the event that the operator is not in the cab?

2.19 How does an OCC to onboard PA announcement route to the onboard communications system?

2.20 How does the OCC send real-time text display information to the LED signs and LCD screens?

2.21 HR and Inter-M needs to confirm the CCTV interface to wayside equipment. Will the recommended NICE software be used or will the system proposed by HR and Inter-M be compatible with the wayside CCTV equipment?

2.22 Please review and respond to the attached DTO document containing communications and CCTV comments and questions.

2.23 Please confirm HR and Inter-M/Woojin attendance for a communications workshop currently scheduled for July 27 in Denver. It would be helpful to have the user configurable decisions matrix submitted to DTS/DTO before that date so the desired functionality can be discussed.

The following are additional comments provided by Ned Parker (RTD Vehicle Engineer) after the PDR meeting. Please review these comments and provide a response.

- Will the onboard CCTV system have coverage of the underseat areas to help detect unattended bags/packages?
- Will the vehicles have microphones to allow real-time monitoring of the in-car noise? The concern is that, in an emergency situation, can the OCC monitor the audio and visual state of the car? OCC can monitor one video feed and can broadcast passenger information through the PA system. Can it listen to passengers in general or only when the ADA/PEI call button is pressed?
    - The PA system has ambient noise-level microphones designed into it. Can these capture discernable voices? Could these be ported to the car-to-

	DTS/DTP Submittal Review Comments Continued: wayside comm system?

Communications PDR
Follow-up questions from DTO

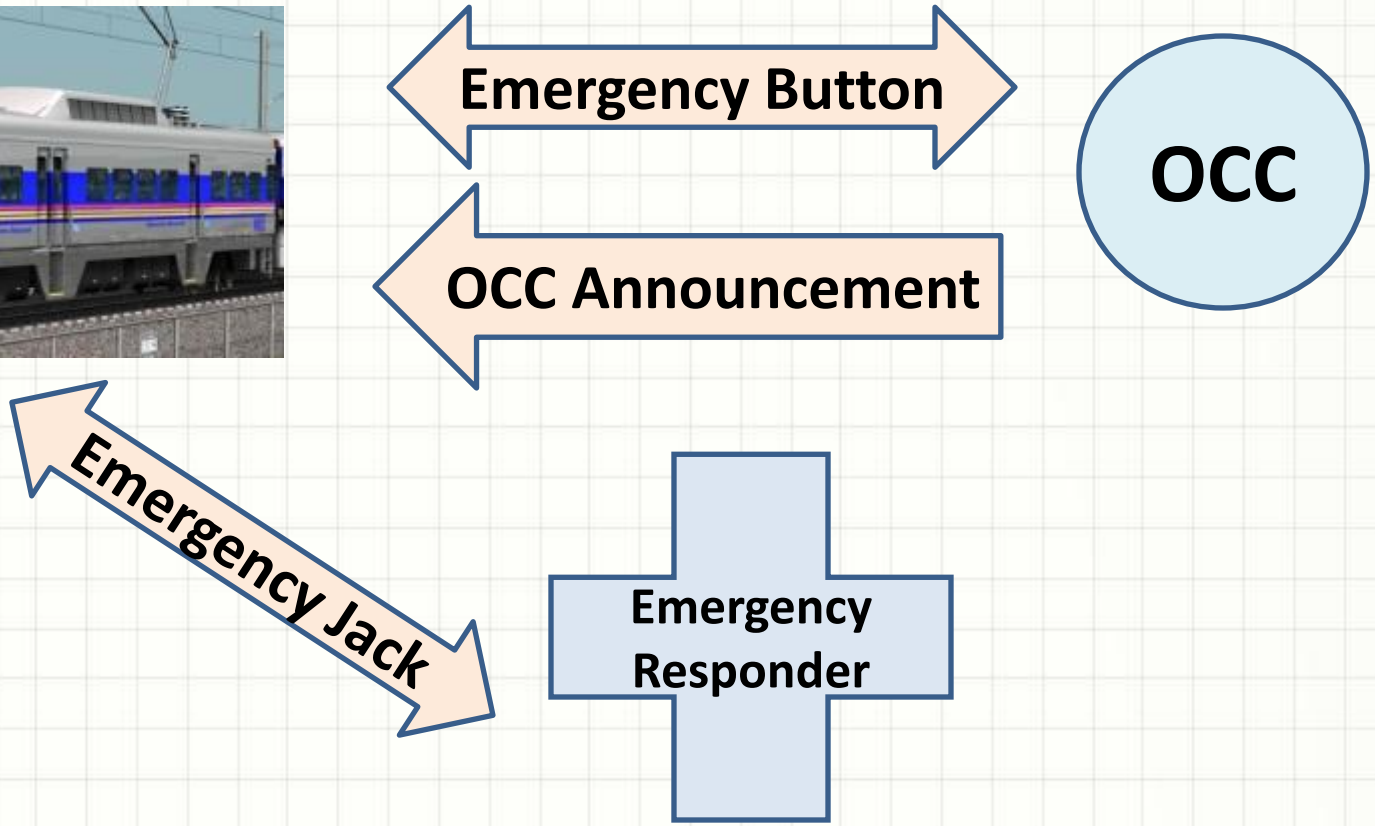
May 25, 2011

Communications Follow-up

PA	• Passenger Communication
VMS	• Passenger Information
MDS	• Vehicle Health Alerts
APC	• Passenger Counts
CCTV	• Video Recording
TEL	• Telephones

PA - Passenger Communication

- Operator makes onboard announcements using PA system.
- Passengers can call the Operator by pushing the Emergency Call button. If the Operator fails to respond, the call will be forwarded to the OCC.
- The OCC also has the ability to make announcements directly to the train.
- The ability for an Emergency Responder to communicate with the passengers from outside the vehicle is also required.



VMS - Passenger Information

How will information content be updated on the vehicle for:

- Destination Signs?
- Next Stop Announcements?
- Video screens with stop and public information?
- Video screens with marketing materials?

VMS Content

CRMF

How do we update the content?

- Will software be provided to capture & send content to vehicle?
- Can content be sent to the entire fleet or a specific subset of trains?
- Can content be loaded before needed and activated at a specific date/time?

MDS – Vehicle Health Data

Full MDS Download

Can we complete a full download, via wireless, for all MDS & Event Log data?

MDS data

Can alerts be downloaded to the network during every turn at DUS?

DUS

Alerts sent via network to CRMF Maintenance System

MDS Data

CRMF

APC – Passenger Counts

APC data will be downloaded daily at the CRMF and sent to RTD via the network.

INIT APC data

CRMF

Can the same wireless method be used to download APC, MDS and VMS information ??

CCTV – Video Recording

- Onboard CCTV is recorded and stored for 72 hours
  Recordings can be downloaded at CRMF or HDD pulled as needed
- Stations CCTV is recorded and stored for at least 30 days
- Can CCTV automatically display the appropriate Station or Onboard camera when an Emergency Button is pressed?

OCC Requests CCTV

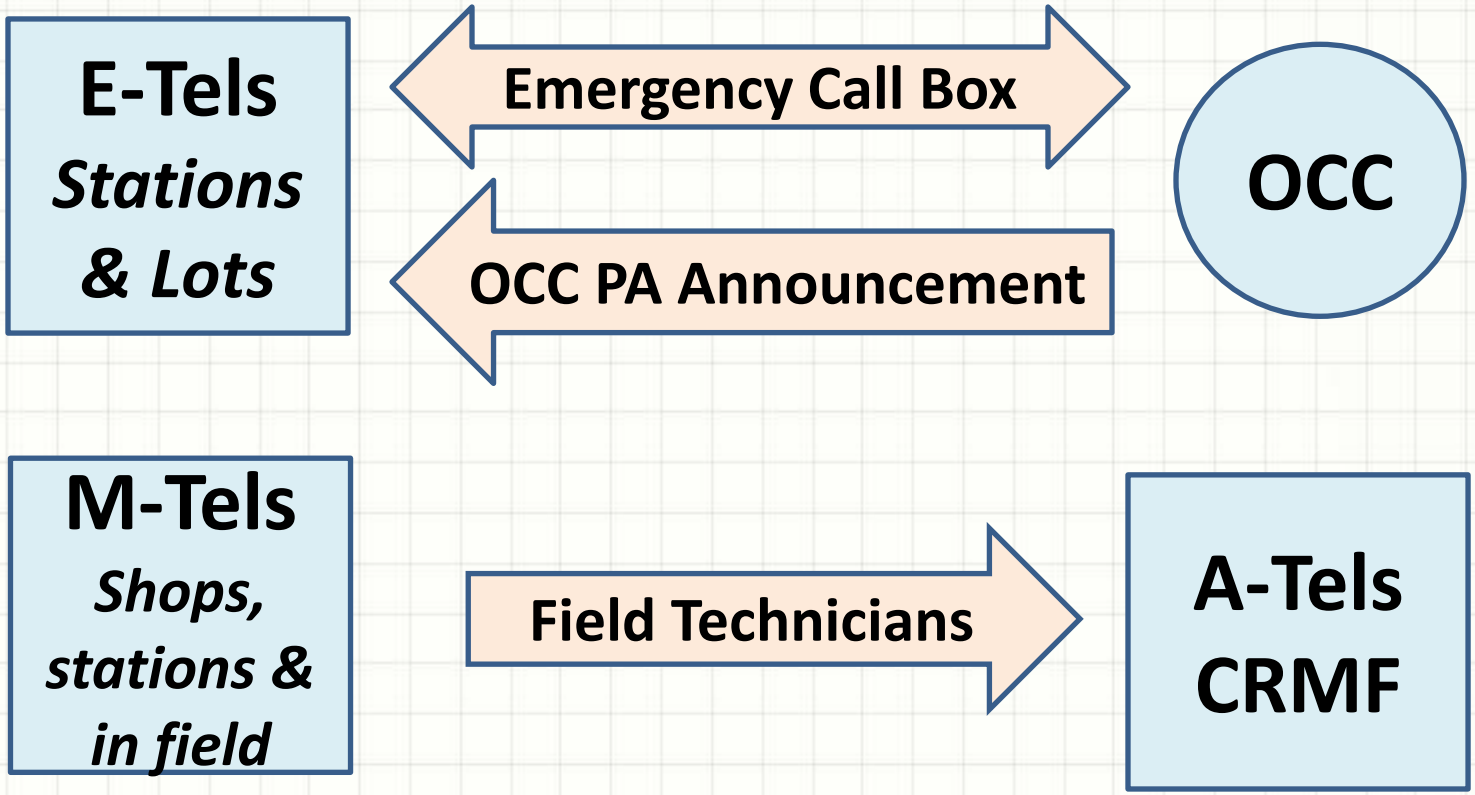
OCC

CCTV downloaded

CRMF

Onboard CCTV

Doors viewed at Platform or Operator selects car & camera.

Telephone Communications

- E-Tels – Emergency Call boxes at stations, tunnels, elevators, etc. Direct dials to the Security Center
- M-Tels – Maintenance call boxes, allow inside 4 digit dialing
- A-Tels – Administrative phone system throughout CRMF, only phones with outside line access.